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A GUIDE DEVICE FOR A SLIDING SCREEN SYSTEM

Abstract

The disclosure relates to a guide device for a sliding screen system. The guide device includes a housing, a height adjustment element arranged to be set in different positions in a height direction, and a manually operable pivot connected to the housing for pivoting around a pivot axis and connected to the height adjust-ment element for moving the height adjustment element when pivoted. The housing includes a first toothed structure extending in a circular path around the pivot axis. The manually operable pivot includes a second toothed structure extending in a circular path around the pivot axis. The first toothed structure and the second toothed structure engage each other. The housing includes at least one finger that is radially flexible relative the pivot axis and that includes a section of the first toothed structure.

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Background/Summary

TECHNICAL FIELD

[0001] The invention relates to a guide device for a sliding screen system comprising a housing, a height adjustment element arranged to be set in different positions in a height direction relative to the housing, a manually operable pivot means pivotably connected to the housing for pivoting around a pivot axis and operatively connected to the height adjustment element for moving the height adjustment element in the height direction when pivoted. The height adjustment element may be provided with a slide rail engagement structure, such as a sliding surface or a roller.

[0002] The invention further relates to a sliding screen system for a sliding screen comprising the guide device and an elongated slide rail, wherein the elongated slide rail and the guide device are adapted for contacting each other during movement of the guide device along the elongated slide rail.

[0003] The sliding screen system may be realized in a piece of furniture, such as a wardrobe, a cabinet, cupboard, sideboard and chest of drawers. Here, a main body (platform) of the piece of furniture is provided with the elongated slide rail and the sliding screen is provided with the guide device, which interact with the elongated slide rail. The sliding screen system may alternatively be realized for a sliding screen that is slidable in relation to a structure, such as a wall/floor/ceiling in a building.

[0004] According to prior art, a main sliding arrangement is adapted to carry a weight of the screen and an auxiliary sliding arrangement is adapted for steering the screen during a sliding movement defined by the main sliding arrangement and reducing the risk that the guide component will fall out of the associated elongated slide rail. The invention is related to such a main sliding arrangement. The main sliding arrangement may be located at a lower edge of the sliding screen so that the sliding screen is then resting on a slide rail (“standing sliding screen”). The auxiliary sliding arrangement is then located at an upper edge of the sliding screen.

[0005] Due to manufacturing tolerances and due to the fact that the floor on which the piece of furniture is placed is often not level, it is usually necessary that the slide rail engagement structure, such as a sliding surface or a roller, is adjustable in its vertical position (i.e. in a height direction) with respect to the sliding screen. Various height adjustment systems are known for this purpose.

[0006] DE 20 2016 102 679 U1 discloses a guide device for a sliding screen comprising a housing, a height adjustment element arranged to be set in different positions in a height direction relative to the housing, a manually operable pivot means pivotably connected to the housing for pivoting around a pivot axis and operatively connected to the height adjustment element for moving the height adjustment element in the height direction when pivoted. The manually operable pivot means has an adjusting wheel with external teeth and a shaft extending from this adjusting wheel. The shaft has a first bearing section, an adjoining intermediate adjustment section and a second bearing section, which are all circular-cylindrical. The first bearing section and the second bearing section being coaxial with one another, so that their common axis defines the pivot axis. The housing comprises two walls spaced in the direction of the pivot axis and each wall being provided with a circular hole for receipt of the first bearing section and the second bearing section, respectively. The intermediate adjustment section is eccentric with respect to the pivot axis. The outer surface of the intermediate adjustment section forms a contact surface for effecting an upwards facing contact surface of the height adjustment element. Further, the housing has two opposing guide surfaces on either side of the pivot axis, which extend in the height direction, for

guiding the height adjustment element. When the housing is mounted on a sliding screen, the contact surfaces are always in contact with each other due to gravitation, since the weight of the sliding screen acts on the housing. Thus, when the adjusting wheel is rotated in one direction, the height adjustment element may be adjusted upwards/downwards.

SUMMARY

[0007] A first object of the invention is to achieve a guide device for a sliding screen, which creates conditions for a more space-efficient solution.

[0008] The object is achieved by a device according to claim 1. Thus, it is achieved by a guide device for a sliding screen comprising a housing, a height adjustment element arranged to be set in different positions in a height direction relative to the housing, a manually operable pivot means pivotably connected to the housing for pivoting around a pivot axis and operatively connected to the height adjustment element for moving the height adjustment element in the height direction when pivoted, characterized in that the manually operable pivot means is adapted to support the height adjustment element so that the height adjustment element may be moved upwards and downwards, respectively, between the different positions in the height direction corresponding to a pivoting of the manually operable pivot means clockwise and counter clockwise, respectively.

[0009] According to one example, each one of the manually operable pivot means and the height adjustment element comprises a bearing section and the bearing sections are adapted for engagement. Thus, the height adjustment element is carried by the manually operable pivot means. According to one example, the height adjustment element is journaled in the manually operable pivot means.

[0010] Since the manually operable pivot means is adapted to support the height adjustment element, a total installation height may be reduced in comparison to prior art. More specifically, the manually operable pivot means may be arranged in relation to the housing such that the pivot axis is at a lower position in the height direction relative to an upper peripheral surface of a main body of the height adjustment element.

[0011] Since the manually operable pivot means is adapted to support the height adjustment element, the housing does not need to be designed for supporting the height adjustment element. Thus, the housing may be of a simple design creating conditions for a cost-efficient manufacturing.

[0012] According to one embodiment example, the height adjustment element is adapted to be moved in an arcuate path relative to the housing around the pivot axis between the different positions in the height direction. The housing may be provided with an opening of a sufficient size to allow for the movement of the height adjustment element in the arcuate path relative to the housing. Accordingly, the housing may be provided with an opening of a sufficient transverse size to allow for free movement of the height adjustment element without contacting the housing in a transverse direction relative to the height direction.

[0013] According to one further development of the last-mentioned embodiment example, the height adjustment element is adapted to be arranged parallel displaced in different height positions along the arcuate path. It is advantageous when the height adjustment element is provided with a slide rail engagement structure in the form of a sliding surface since the orientation of the sliding surface may be the same irrespective of the height position of the height adjustment element.

[0014] According to one further embodiment example, a first one of the manually operable pivot means and the height adjustment element comprises a projection extending in parallel with the pivot axis, wherein an outer peripheral surface of the projection defines a first engagement structure, and wherein a second one of the manually operable pivot means and the height adjustment element comprises a recess extending in parallel with the pivot axis, wherein an inner peripheral surface defining the recess defines a second engagement structure and wherein the projection is adapted to be received in the recess so that the first engagement structure and the second engagement structure engage each other.

[0015] According to one example, the manually operable pivot means comprises a projection in the

form of a shaft and the height adjustment element comprises a recess in the form of a hole, wherein the shaft and the hole are provided with substantially the same shape and size.

[0016] According to one further development of the last-mentioned embodiment example, each one of the projection and the recess has a circular cylindrical shape for allowing a relative pivoting movement between the manually operable pivot means and the height adjustment element. Thus, the circular cylindrical portions of the manually operable pivot means and the height adjustment element forms bearing sections adapted for engagement with each other.

[0017] According to one further development of the last-mentioned embodiment example, a centre axis of the projection or recess of the manually operable pivot means is in parallel with and offset in relation to the pivot axis. In other words, the centre axis of the projection or recess of the manually operable pivot means is eccentric in relation to the pivot axis. Due to the eccentricity, the height adjustment element may be moved in the arcuate path.

[0018] According to one further embodiment example, the housing comprises a through hole adapted to receive a portion of the manually operable pivot means, wherein the through hole extends in a transverse direction relative to the height direction, wherein the manually operable pivot means comprises a first part adapted to be moved into the through hole from a first end of the through hole and a second part adapted to be moved into the through hole from a second end of the through hole and wherein the first part and the second part are adapted for being rotationally rigidly connected to each other via a relative motion in parallel with an axis of said through hole. It creates conditions for an easy assembly.

[0019] According to one further development of the last-mentioned embodiment example, the first part comprises a handle for being pivoted by a user for pivoting the manually operable pivot means relative to the housing. It creates further conditions for an easy assembly.

[0020] According to one further development of the last-mentioned embodiment example, the second part comprises the projection or recess provided with the first engagement structure or second engagement structure. Thus, the second part, which is assembled from a back side of the housing relative to the first part with the handle, may be provided with the bearing section for engagement with the height adjustment element.

[0021] According to one further development of the last-mentioned embodiment example, the first part and the second part comprises snap engagement structures for a snapping engagement during the relative motion in parallel with the axis of said through hole. It creates further conditions for an easy assembly.

[0022] According to one further embodiment example, the housing comprises a first toothed structure extending in a circular path around the pivot axis, wherein the manually operable pivot means comprises a second toothed structure extending in a circular path around the pivot axis, wherein the manually operable pivot means is adapted to be arranged relative to the housing so that the first toothed structure and the second toothed structure engage each other, wherein the toothed structures define a plurality of discrete relative angular positions, which correspond to the different positions of the height adjustment element in the height direction relative to the housing.

[0023] According to one further development of the last-mentioned embodiment example, the housing comprises at least one finger that is radially flexible relative the pivot axis and that comprises a section of the first toothed structure and is adapted to flex radially outwards during pivoting when opposite teeth meet and back again when the teeth of one of the first toothed structure and the second toothed structure meet spaces between the teeth of the other one of the first toothed structure and the second toothed structure during height adjustment.

[0024] According to one further embodiment example, the manually operable pivot means is arranged relative to the housing in a way that the pivot axis is perpendicular relative to the height direction.

[0025] According to one further embodiment example, the height adjustment element is adapted for being guided by an associated elongated slide rail during movement of the guide device relative to

the elongated slide rail and wherein the height adjustment regards adjustment in a direction perpendicular to the elongated slide rail.

[0026] According to one further development of the last-mentioned embodiment example, the height adjustment element comprises a first slide surface that is adapted for a sliding contact with a second slide surface of the elongated slide rail during the movement of the guide device relative to the elongated slide rail.

[0027] According to a further aspect of the invention, it regards a sliding screen system for a sliding screen, wherein the sliding screen system comprises the guide device according to any preceding embodiment example, and an elongated slide rail, wherein the elongated slide rail and the guide device are adapted for contacting each other during movement of the guide device along the elongated slide rail.

[0028] According to one embodiment example, the sliding screen system comprises a sliding screen, a platform and a guide device according to any one of the above embodiment examples, wherein the housing is adapted for being rigidly connected to the sliding screen and the elongated slide rail is adapted for being rigidly connected to the platform so that the sliding screen may be moved relative to the platform in a way that the height adjustment element is in engagement with the elongated slide rail.

[0029] The sliding screen system may be realized in a piece of furniture, such as a wardrobe, a cabinet, cupboard, sideboard and chest of drawers. Here, a main body (platform) of the piece of furniture, such as a panel, is provided with the elongated slide rail and the sliding screen is provided with the guide device, which interacts with the elongated slide rail. The sliding screen system may alternatively be realized for a sliding screen that is slidable in relation to a structure, such as a wall/floor/ceiling in a building.

[0030] According to one further embodiment example, the guide device is arranged at a lower end of the sliding screen in the vertical direction and adapted to carry the weight of the sliding screen.

[0031] Further advantages and advantageous features of the invention are disclosed in the following description and in the dependent claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] With reference to the appended drawings, below follows a more detailed description of embodiments of the invention cited as examples.

[0033] In the drawings:

[0034] FIG. 1A is a perspective exploded view of a guide device according to a first embodiment in a front view,

[0035] FIG. 1B is a perspective exploded view of the guide device in FIG. 1A in a rear view,

[0036] FIG. 2 is a side view of parts of the guide device in FIG. 1A,

[0037] FIG. 3A, FIG. 3B and FIG. 3C indicate operations for different height positions of a height adjustment element in the guide device in FIG. 1A,

[0038] FIG. 4 is a perspective view of a sliding screen system for a sliding screen comprising the guide device in FIG. 1A, and

[0039] FIG. 5 is a partly cut perspective view of the sliding screen system according to FIG. 4.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0040] FIG. 1A is a perspective exploded view of a guide device 2 for a sliding screen according to a first embodiment in a front view and FIG. 1B is a corresponding view of the guide device 2 in a rear view.

[0041] The guide device 2 comprises a housing 4, a height adjustment element 6 arranged to be set in different positions in a height direction H relative to the housing 4. The housing 4 comprises a

receptacle **8** for receipt of the height adjustment element **6**. More specifically, the housing **4** comprises a first wall **10** and a second wall **12** spaced from the first wall **10** so that the receptacle is formed between the walls **10**, **12**.

[0042] The guide device **2** further comprises a manually operable pivot means **14** pivotably connected to the housing **4** for pivoting around a pivot axis P and operatively connected to the height adjustment element **6** for moving the height adjustment element in the height direction when pivoted.

[0043] The first wall **10** of the housing **4** is provided with a first through hole **16** and the second wall **12** of the housing **4** is provided with a second through hole **18**. The first through hole **16** and the second through hole **18** are arranged in line with each other forming a common through hole of the housing **4**. The manually operable pivot means **14** comprises a first part **20** adapted to be moved into the first through hole **16** from a first end of the common through hole and a second part **21** adapted to be moved into the second through hole **18** from a second end of the common through hole. The first part **20** and the second part **21** are adapted for being rotationally rigidly connected to each other via a relative motion in parallel with an axis of said through holes **16**, **18**.

[0044] The second wall **12** of the housing **4** has an inner peripheral surface **23** around the second through hole **18**. The circular inner peripheral surface **23** defines a bearing engagement structure. The second part **21** of the manually operable pivot means **14** comprises a corresponding outer peripheral surface **25** defining a corresponding bearing engagement structure. The second part **21** of the manually operable pivot means **14** is adapted to be received in the second through hole **18** so that the engagement structures engage each other. More specifically, each one of the inner peripheral surface **23** and the outer peripheral surface **25** has a circular cylindrical shape for allowing a relative pivoting movement between the manually operable pivot means **14** and the second wall **12**.

[0045] The first part **20** comprises a handle **22** for being pivoted by a user for pivoting the manually operable pivot means **14** relative to the housing **4**. The first part **20** further comprises a recess **24** having a first engagement structure **26**. The second part **21** comprises a projection **28** having a second engagement structure **30**. More specifically, the first engagement structure **26** and the second engagement structure **30** are adapted for a snapping engagement during the relative motion in parallel with the axis of said through hole. More specifically, each one of the recess **24** and the projection **28** is provided with a non-circular engagement surface. More specifically, each one of the recess **24** and the projection **28** is provided with a rectangular engagement surface. Further, the projection **28** is provided with two spaced flexible fingers, which are adapted to be received in the recess **24**.

[0046] The height adjustment element **6** comprises a through hole **32**, which is adapted to receive a portion of the manually operable pivot means **14**. The manually operable pivot means **14** is adapted to support the height adjustment element **6** so that the height adjustment element may be moved upwards and downwards, respectively, between the different positions in the height direction corresponding to a pivoting of the manually operable pivot means clockwise and counter clockwise, respectively.

[0047] The projection **28** of the second part **21** of the manually operable pivot means **14** has an outer peripheral surface **34** which defines a first bearing engagement structure and the height adjustment element **6** comprises a recess **32**, which is formed by said through hole, extending in parallel with the pivot axis, wherein an inner peripheral surface **36** defining the recess **32** defines a second bearing engagement structure and wherein the projection **28** is adapted to be received in the recess **32** so that the first engagement structure and the second engagement structure engage each other. More specifically, each one of the projection **28** and the recess **32** has a circular cylindrical shape for allowing a relative pivoting movement between the manually operable pivot means **14** and the height adjustment element **6**. Further, a centre axis **38** of the projection **28** of the manually operable pivot means **14** is in parallel with and offset in relation to the pivot axis P, see FIG. 2.

Whilst the second part **21** of the manually operable pivot means **14**, in particular the outer peripheral surface **34**, is shown having a substantially circular profile, in some embodiments it may have a substantially square profile, in particular at its base.

[0048] The housing **4** comprises a first toothed structure **48** extending in a circular path around the pivot axis P. Further, the manually operable pivot means **14** comprises a second toothed structure **50** extending in a circular path around the pivot axis, wherein the manually operable pivot means **14** is adapted to be arranged relative to the housing so that the first toothed structure **48** and the second toothed structure **50** engage each other, wherein the toothed structures **48**, **50** define a plurality of discrete relative angular positions, which correspond to the different positions of the height adjustment element **6** in the height direction relative to the housing **4**. In some embodiments, the spacing of the teeth may be irregular. For example, the first toothed structure **48** may have fewer teeth (i.e. a larger gap between angular positions) at the bottom of the housing **4** than at the top.

[0049] More specifically, the housing **4** comprises one or more radially flexible fingers **52**, **54**. In the example show, the housing **4** comprises two spaced fingers **52**, **54**, one on each side of the pivot axis P, that are radially flexible relative the pivot axis P towards and away from each other. Each finger **52**, **54** comprises a section of the first toothed structure **48** and is adapted to flex radially outwards during pivoting when opposite teeth meet and back again when the teeth of one of the first toothed structure **48** and the second toothed structure **50** meet spaces between the teeth of the other one of the first toothed structure and the second toothed structure during height adjustment. Thus, the first toothed structure **48** is non-continuous/interrupted in the circumferential direction. The purpose of the fingers **52**, **54** is to allow the handle **22** to be turned between the different distinct height positions using a suitable force. In other words, if there were no flexible fingers, it would require more force to move the handle **22**. Furthermore, the teeth will be experience less wear in the presence of the flexible fingers **52**, **54** relative to the case of a rigid structure.

[0050] More specifically, the manually operable pivot means **14** comprises a projection that comprises the second toothed structure **50**. The second toothed structure **48** is continuous in the circumferential direction.

[0051] Further, the manually operable pivot means **14** is arranged relative to the housing **4** in a way that the pivot axis P is perpendicular relative to the height direction H.

[0052] Further, the housing **4** comprises a safety portion **51** in the form of a fin adapted for being received in an elongated slide rail **56**, see FIG. 5, for reducing the risk that the sliding screen will be displaced transversally relative to the elongated slide rail.

[0053] FIG. 2 is a side view of parts of the guide device **2** in FIG. 1A. More specifically, the arrangement and operation of the height adjustment element **6** and the manually operable pivot means **14** is disclosed in more detail. Movement of the handle **22** around the pivot axis P is indicated via arrows **40**, **42**. Such a movement of the handle **22** results in a height adjustment of the height adjustment element **6**, see arrows **44**, **46**.

[0054] More specifically, the height adjustment element **6** is adapted to be moved in an arcuate path **60** relative to the housing **4** around the pivot axis P between the different positions in the height direction. More specifically, the height adjustment element **6** is adapted to be arranged parallel displaced in different height positions along the arcuate path. It will be shown in more detail in FIG. 3A, FIG. 3B and FIG. 3C.

[0055] The height adjustment element **6** is adapted for being guided by an associated elongated slide rail **56** (see FIG. 4) during movement of the guide device **2** relative to the elongated slide rail and wherein the height adjustment regards adjustment in a direction perpendicular to the elongated slide rail. Further, the height adjustment element **6** comprises a first slide surface **58** that is adapted for a sliding contact with a second slide surface of the elongated slide rail **56** during the movement of the guide device **2** relative to the elongated slide rail.

[0056] FIG. 3A, FIG. 3B and FIG. 3C indicate operations for different height positions of the

height adjustment element **6** in the guide device **2** in FIG. **1A**. More specifically, FIG. **3A** indicates the handle **22** of the manually operable pivot means **14** being moved to an end stop in the counterclockwise direction defining a maximally withdrawn position of the height adjustment element **6**, see minimum height $h_{sub.min}$. In a similar manner, FIG. **3C** indicates the handle **22** of the manually operable pivot means **14** being moved to an end stop in the clockwise direction defining a maximally extended position of the height adjustment element **6**, see maximum height $h_{sub.max}$. FIG. **3B** indicates the handle **22** of the manually operable pivot means **14** being moved to a center intermediate position in the clockwise direction between the end stop positions defining a nominal height position of the height adjustment element **6**.

[0057] FIG. **4** is a perspective view of a sliding screen system **64** for a sliding screen **66** comprising the guide device **2** in FIG. **1A** and the elongated slide rail **56**, wherein the elongated slide rail **56** and the guide device **2** are adapted for contacting each other during movement of the guide device along the elongated slide rail. The sliding screen system further comprises the sliding screen **66** and a platform **68**, wherein the housing **4** is adapted for being rigidly connected to the sliding screen **66**, see FIG. **5**, and the elongated slide rail **56** is adapted for being rigidly connected to the platform **68** so that the sliding screen may be moved relative to the platform in a way that the height adjustment element **6** is in engagement with the elongated slide rail. The guide device **2** is arranged at a lower end of the sliding screen **66** in the vertical direction and adapted to carry the weight of the sliding screen.

[0058] The safety portion **51** of the housing **4** is received in the elongated slide rail **56** in a way that a free end of the safety portion is positioned at a distance from a bottom surface of the slide rail. Further, the sliding screen is arranged in such a way relative to the elongated slide rail **56** that the safety portion **51** of the housing **4** is received in the elongated slide rail **56** in a way that opposite contact surfaces of the safety portion **51** are spaced from the associated contact surfaces of the walls of the elongated slide rail **56**. In other words, the guide portion **51** of the housing **4** runs free from contact with the elongated slide rail **56** in a normal state.

[0059] Also disclosed are examples according to the following clauses: [0060] 1. A guide device (**2**) for a sliding screen system comprising a housing (**4**), a height adjustment element (**6**) arranged to be set in different positions in a height direction (H) relative to the housing, a manually operable pivot means (**14**) pivotably connected to the housing (**4**) for pivoting around a pivot axis (P) and operatively connected to the height adjustment element (**6**) for moving the height adjustment element (**6**) in the height direction when pivoted, wherein the manually operable pivot means (**14**) is adapted to support the height adjustment element (**6**) so that the height adjustment element may be moved upwards and downwards, respectively, between the different positions in the height direction (H) corresponding to a pivoting of the manually operable pivot means (**14**) clockwise and counter clockwise, respectively. [0061] 2. A guide device according to clause 1, wherein the height adjustment element (**6**) is adapted to be moved in an arcuate path (**60**) relative to the housing (**4**) around the pivot axis (P) between the different positions in the height direction (H). [0062] 3. A guide device according to clause 2, wherein the height adjustment element (**6**) is adapted to be arranged parallel displaced in different height positions along the arcuate path. [0063] 4. A guide device according to any preceding clause, wherein a first one of the manually operable pivot means (**14**) and the height adjustment element (**6**) comprises a projection (**28**) extending in parallel with the pivot axis, wherein an outer peripheral surface (**34**) of the projection defines a first engagement structure, and wherein a second one of the manually operable pivot means (**14**) and the height adjustment element (**6**) comprises a recess (**32**) extending in parallel with the pivot axis, wherein an inner peripheral surface (**36**) defining the recess defines a second engagement structure and wherein the projection is adapted to be received in the recess so that the first engagement structure and the second engagement structure engage each other. [0064] 5. A guide device according to clause 4, wherein each one of the projection (**28**) and the recess (**32**) has a circular cylindrical shape for allowing a relative pivoting movement between the manually operable pivot means and the

height adjustment element. [0065] 6. A guide device according to clause 4 or 5, wherein a centre axis (38) of the projection (28) or recess (32) of the manually operable pivot means (14) is in parallel with and offset in relation to the pivot axis (P). [0066] 7. A guide device according to any preceding clause, wherein the housing (4) comprises a through hole (16, 18) adapted to receive a portion of the manually operable pivot means (14), wherein the through hole extends in a transverse direction relative to the height direction (H), wherein the manually operable pivot means (14) comprises a first part (20) adapted to be moved into the through hole from a first end of the through hole and a second part (21) adapted to be moved into the through hole from a second end of the through hole and wherein the first part and the second part are adapted for being rotationally rigidly connected to each other via a relative motion in parallel with an axis of said through hole. [0067] 8. A guide device according to clause 7, wherein the first part (20) comprises a handle (22) for being pivoted by a user for pivoting the manually operable pivot means relative to the housing. [0068] 9. A guide device according to clause 7 or 8, wherein the second part (21) comprises the projection (28) or recess provided with the first engagement structure or second engagement structure. [0069] 10. A guide device according to any one of clauses 7-9, wherein the first part (20) and the second part (21) comprises snap engagement structures (26, 30) for a snapping engagement during the relative motion in parallel with the axis of said through hole. [0070] 11. A guide device according to any preceding clause, wherein the housing (4) comprises a first toothed structure (48) extending in a circular path around the pivot axis (P), wherein the manually operable pivot means (14) comprises a second toothed structure (50) extending in a circular path around the pivot axis (P), wherein the manually operable pivot means is adapted to be arranged relative to the housing so that the first toothed structure and the second toothed structure engage each other, wherein the toothed structures define a plurality of discrete relative angular positions, which correspond to the different positions of the height adjustment element in the height direction relative to the housing. [0071] 12. A guide device according to clause 11, wherein the housing (4) comprises at least one finger (52) that is radially flexible relative the pivot axis (P) and that comprises a section of the first toothed structure (48) and is adapted to flex radially outwards during pivoting when opposite teeth meet and back again when the teeth of one of the first toothed structure and the second toothed structure meet spaces between the teeth of the other one of the first toothed structure and the second toothed structure during height adjustment. [0072] 13. A guide device according to any preceding clause, wherein the manually operable pivot means (14) is arranged relative to the housing (4) in a way that the pivot axis (P) is perpendicular relative to the height direction (H). [0073] 14. A guide device according to any preceding clause, wherein the guide device (2) is adapted for being guided by an associated elongated slide rail (56) during movement of the guide device (2) relative to the elongated slide rail (56) and wherein the height adjustment regards adjustment in a direction perpendicular to the elongated slide rail. [0074] 15. A guide device according to clause 14, wherein the height adjustment element (4) comprises a first slide surface (58) that is adapted for a sliding contact with a second slide surface of the elongated slide rail during the movement of the guide device relative to the elongated slide rail. [0075] 16. A sliding screen system (64) for a sliding screen (66), wherein the sliding screen system comprises the guide device (2) according to any preceding clause, and an elongated slide rail (56), wherein the elongated slide rail and the guide device are adapted for contacting each other during movement of the guide device along the elongated slide rail. [0076] 17. A sliding screen system (64) according to clause 16, wherein the sliding screen system comprises a sliding screen (66), a platform (68) and a guide device (2) according to any one of clauses 1-15, wherein the housing (4) is adapted for being rigidly connected to the sliding screen (66) and the elongated slide rail (56) is adapted for being rigidly connected to the platform (68) so that the sliding screen may be moved relative to the platform in a way that the height adjustment element is in engagement with the elongated slide rail. [0077] 18. A sliding screen system (64) according to clause 16 or 17, wherein the guide device (2) is arranged at a lower end of the sliding screen (66) in the vertical direction and adapted to carry

the weight of the sliding screen.

[0078] It is to be understood that the present invention is not limited to the embodiments described above and illustrated in the drawings; rather, the skilled person will recognize that many changes and modifications may be made within the scope of the appended claims.

Claims

1. A guide device comprising: a housing; a height adjustment element arranged to be set in different positions in a height direction relative to the housing; a manually operable pivot pivotably connected to the housing for pivoting around a pivot axis and operatively connected to the height adjustment element for moving the height adjustment element in the height direction when pivoted, wherein the manually operable pivot is adapted to support the height adjustment element so that the height adjustment element may be moved upwards and downwards, respectively, between the different positions in the height direction corresponding to a pivoting of the manually operable pivot clockwise and counter clockwise, respectively, wherein the housing comprises a first toothed structure extending in a first circular path around the pivot axis, wherein the manually operable pivot comprises a second toothed structure extending in a second circular path around the pivot axis, wherein the manually operable pivot is adapted to be arranged relative to the housing so that the first toothed structure and the second toothed structure engage each other, wherein the toothed structures define a plurality of discrete relative angular positions, which correspond to the different positions of the height adjustment element in the height direction relative to the housing, wherein the housing comprises at least one finger that is radially flexible relative the pivot axis and that comprises a section of the first toothed structure and is adapted to flex radially outwards during pivoting when opposite teeth meet and back again when the teeth of one of the first toothed structure and the second toothed structure meet spaces between the teeth of the other one of the first toothed structure and the second toothed structure during height adjustment.
2. The guide device of to claim 1, wherein the height adjustment element is adapted to be moved in an arcuate path relative to the housing around the pivot axis between the different positions in the height direction.
3. The guide device of to claim 2, wherein the height adjustment element is adapted to be arranged parallel displaced in different height positions along the arcuate path.
4. The guide device of claim 1, wherein; a first one of the manually operable pivot and the height adjustment element comprises a projection extending in parallel with the pivot axis, an outer peripheral surface of the projection defines a first engagement structure, a second one of the manually operable pivot and the height adjustment element comprises a recess extending in parallel with the pivot axis, an inner peripheral surface defining the recess defines a second engagement structure, and the projection is adapted to be received in the recess so that the first engagement structure and the second engagement structure engage each other.
5. The guide device of to claim 4, wherein each one of the projection and the recess has a circular cylindrical shape for allowing a relative pivoting movement between the manually operable pivot and the height adjustment element.
6. The guide device of to claim 4, wherein a centre axis of the projection or the recess of the manually operable pivot is in parallel with and offset in relation to the pivot axis.
7. The guide device of 4, wherein: the housing comprises a through hole adapted to receive a portion of the manually operable pivot, the through hole extends in a transverse direction relative to the height direction, the manually operable pivot comprises a first part adapted to be moved into the through hole from a first end of the through hole and a second part adapted to be moved into the through hole from a second end of the through hole, and the first part and the second part are adapted for being rotationally rigidly connected to each other via a relative motion in parallel with an axis of said through hole.

8. The guide device of to claim 7, wherein the first part comprises a handle for being pivoted by a user for pivoting the manually operable pivot relative to the housing.
9. The guide device of to claim 7, wherein the second part comprises the projection or the recess provided with the first engagement structure or the second engagement structure.
10. The guide device of claim 7, wherein the first part and the second part comprise snap engagement structures for a snapping engagement during the relative motion in parallel with the axis of said through hole.
11. The guide device of claim 1, wherein the manually operable pivot is arranged relative to the housing in a way that the pivot axis is perpendicular relative to the height direction.
12. The guide device of claim 1, wherein; the guide device is adapted for being guided by an associated elongated slide rail during movement of the guide device relative to the elongated slide rail, and the height adjustment regards adjustment in a direction perpendicular to the elongated slide rail.
13. The guide device of claim 12, wherein the height adjustment element comprises a first slide surface that is adapted for a sliding contact with a second slide surface of the elongated slide rail during the movement of the guide device relative to the elongated slide rail.
14. A sliding screen system comprising: a guide device; and an elongated slide rail, wherein the elongated slide rail and the guide device are adapted for contacting each other during movement of the guide device along the elongated slide rail, and wherein the guide device comprises: a housing; a height adjustment element arranged to be set in different positions in a height direction relative to the housing; a manually operable pivot pivotably connected to the housing for pivoting around a pivot axis and operatively connected to the height adjustment element for moving the height adjustment element in the height direction when pivoted, wherein the manually operable pivot is adapted to support the height adjustment element so that the height adjustment element may be moved upwards and downwards, respectively. between the different positions in the height direction corresponding to a pivoting of the manually operable pivot clockwise and counter clockwise, respectively, wherein the housing comprises a first toothed structure extending in a first circular path around the pivot axis, wherein the manually operable pivot comprises a second toothed structure extending in a second circular path around the pivot axis, wherein the manually operable pivot is adapted to be arranged relative to the housing so that the first toothed structure and the second toothed structure engage each other, wherein the toothed structures define a plurality of discrete relative angular positions, which correspond to the different positions of the height adjustment element in the height direction relative to the housing, and wherein the housing comprises at least one finger that is radially flexible relative the pivot axis and that comprises a section of the first toothed structure and is adapted to flex radially outwards during pivoting when opposite teeth meet and back again when the teeth of one of the first toothed structure and the second toothed structure meet spaces between the teeth of the other one of the first toothed structure and the second toothed structure during height adjustment.
15. The sliding screen system of claim 14, further comprising: a sliding screen; and a platform, wherein the housing is adapted for being rigidly connected to the sliding screen and the elongated slide rail is adapted for being rigidly connected to the platform so that the sliding screen may be moved relative to the platform in a way that the height adjustment element is in engagement with the elongated slide rail.
16. The sliding screen system claim 15, wherein the guide device is arranged at a lower end of the sliding screen in a vertical direction and adapted to carry a weight of the sliding screen.
17. The guide device of claim 5, wherein a centre axis of the projection or the recess of the manually operable pivot is in parallel with and offset in relation to the pivot axis.
18. The guide device of claim 8, wherein the second part comprises the projection or the recess provided with the first engagement structure or the second engagement structure.
19. The guide device of claim 8, wherein the first part and the second part comprise snap

engagement structures for a snapping engagement during the relative motion in parallel with the axis of said through hole.

20. The guide device of claim 9, wherein the first part and the second part comprise snap engagement structures for a snapping engagement during the relative motion in parallel with the axis of said through hole.
